

Machine learning for intelligent fiber sensors: from physical design optimization to advanced sensing applications

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Abstract

Fiber sensors encompass optical-fiber platforms and electrically functional fibers, yarns, and textiles that produce spectral, temporal, and spatial data across wearable, biomedical, robotic, infrastructure, and environmental applications. Their responses are shaped by nonlinear transduction, cross-sensitivity, conformal contact, manufacturing variation, drift, and limited calibration coverage. This review examines how machine learning (ML) can be used across the complete fiber-sensing pipeline, from learning-assisted physical design and fabrication-aware optimization to denoising, demodulation, representation learning, multimodal inference, and application-level decision making. Direct demonstrations in optical and electrically functional fiber systems are distinguished from transferable methods developed in adjacent photonic, additive manufacturing, tactile, and soft-sensor platforms. The available evidence is strong for calibration, signal recovery, and task inference within a defined device and operating domain. By contrast, experimental evidence for ML-guided physical co-design, cross-device and cross-batch transfer, long-term stability, and hardware-constrained deployment remains limited. Accordingly, ML is therefore promising as part of a coupled sensor-model-hardware system that preserves task-relevant information and reports when an input lies outside its validated operating range. It is then suggested that validation boundaries rather than headline accuracy alone and identifies uncertainty quantification, physics-guided learning, fabrication metadata, and edge-aware co-design as priorities.

Keywords: fiber sensors; functional fibers; machine learning; inverse design; wearable sensors; distributed sensing

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1. Introduction

Fiber-based sensing is becoming an enabling technology for intelligent monitoring across machines, human body, and infrastructure. Compared with conventional rigid or planar sensors, fiber-based sensors can be more readily routed along curved or deformable surfaces and integrated into soft systems like textiles, while also enabling lightweight, multiplexed, or distributed measurements for both localized sensing and long-range monitoring [1–4]. In wearable, biomedical, and robotic systems, fiber sensors have been integrated into gloves for gesture and grasp recognition [5,6], soft tactile skins

for contact-force and contact-location estimation [7], optical-fiber smartwatches for continuous blood-pressure monitoring [8], and multicore-fiber catheters for continuous three-dimensional shape reconstruction [9]. Across the built and natural environments, distributed fibers sensors are able to support pipeline intrusion monitoring [10,11], railway and bridge structural-health monitoring [12–14], and earthquake or microseismic-event detection over kilometer-scale sensing paths [15,16].

Shown in Figure 1, fiber-based sensing platforms can be classified according to their transduction mechanisms and system-integration routes into two non-exclusive families: electrically functional fibers and optical-fiber sensors. Electrically functional fibers and yarns employ conductive composites [3], piezoelectric materials [17,18], triboelectric structures [19–22], liquid conductors [23–25], or percolating fibrous networks [2,3,17]. These material systems enable integration into garments and smart textiles such as sensing gloves [5,6], plantar-pressure devices [26], electronic skins [27,28], and soft robotic interfaces [29,30]. On the other hand, optical-fiber sensors encode physical or biochemical perturbations through fiber Bragg gratings (FBGs) [8,9], interferometric or intensity-based configurations [31–33], and distributed Rayleigh, Raman, or Brillouin scattering [34–36]. These platforms enable localized or spatially resolved measurements of strain and temperature [35,37–39], pressure and contact force [7,8,40,41], vibration [15,16,42,43], and refractive-index or biochemical changes [44]. For instance, distributed acoustic sensing (DAS) exemplifies optical sensing by using a long optical fiber as a dense sequence of vibration-sensitive channels [4,10,15,16,43]. Despite their different transduction mechanisms, both families can generate high-dimensional spectra [34,37,38,44], time-distance traces [15,16,43], coupled material responses [7,8,39], and spatially correlated array data [7,45,46].

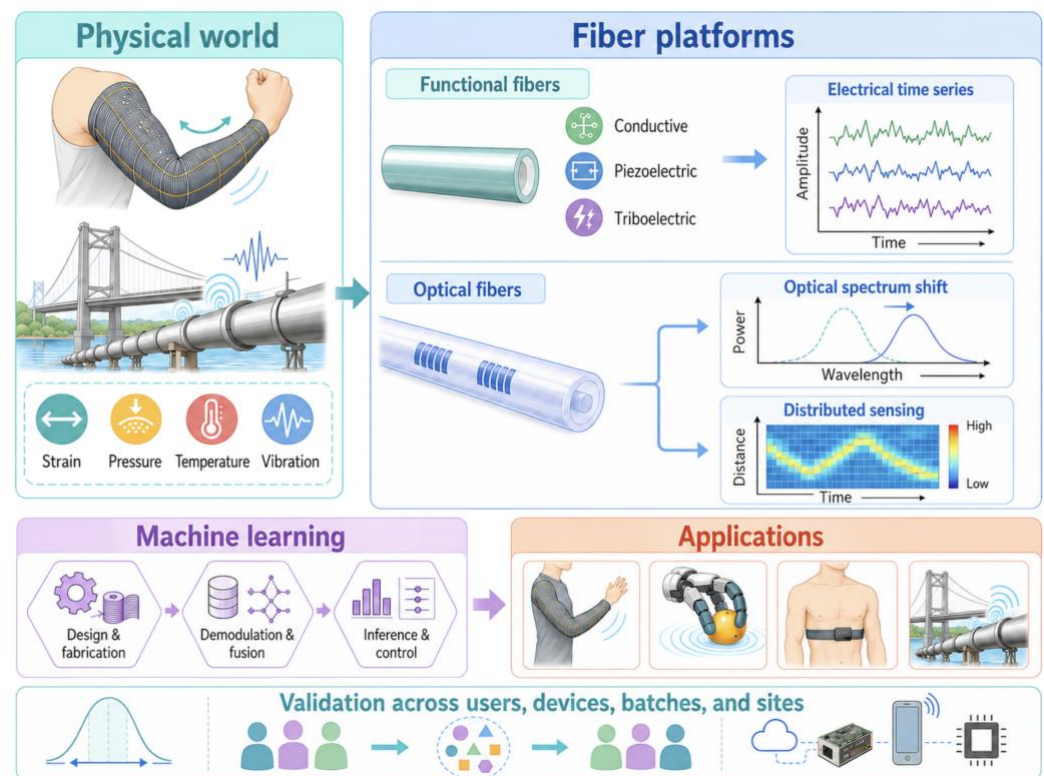


Figure 1. Conceptual framework for machine-learning-enabled intelligent fiber sensing.

The convergence of electrically functional fibers and optical-fiber sensors with multiplexed or distributed readout shifts the central task from detecting a stimulus to

extracting reliable physical information from increasingly complex sensor responses. Admittedly, analytical calibration remains effective when transduction is approximately linear, boundary conditions are controlled, and nuisance variables can be isolated. Complementary data-driven approaches become valuable when conformal contact, material hysteresis, environmental interference, cross-sensitivity, microscopic defects, and manufacturing variability produce nonlinear or non-unique relationships between the physical state and the measured signal [1,3,47]. Hence, the value of machine learning (ML) is not tied to one transduction mechanism, but to where ambiguity enters the sensing pipeline. In fact, ML can contribute at several stages of this information pathway. At the design stage, ML-assisted optimization can guide device geometry, sensor placement, and interrogation-code selection to improve sensitivity, signal separability, or robustness [36,48–57]. During calibration and demodulation, ML models can compensate for drift and device-dependent variation, recover overlapping FBG wavelengths, estimate Brillouin frequency shifts, and suppress noise in distributed measurements [9,34,35,37,38,43,44,58–65]. At the application stage, temporal, spatial, and multimodal models can translate calibrated sensor responses into physical information such as gesture and grasp patterns [5,6], contact-force and contact-location features [7], catheter-shape features [9], physiological signals [8], and infrastructure-event classes [11,15,16]. Collectively, these capabilities position ML as a computational link between physical sensor design, signal interpretation, and application-level decision-making.

1.1. Evolution of fiber sensors

Fiber sensors have evolved from largely uniform waveguides or conductive filaments into hierarchical devices in which geometry, material interfaces, and network organization participate in transduction. Conventional optical fibers are commonly produced by drawing a glass or polymer preform, followed by processes such as grating inscription, tapering, micromachining, surface functionalization, and coating to introduce or modify their sensing functions [33,66–68]. Electrically functional fibers are typically formed through wet spinning, melt spinning, extrusion, coating, or the incorporation of conductive and stimulus-responsive fillers, and can subsequently be woven, knitted, braided, or embroidered into larger sensing structures [2,23–25,69]. These methods offer scalable and repeatable production, while the resulting structures are generally determined by the initial preform, die geometry, coating conditions, or textile assembly pattern.

Thermal drawing has extended this design space by co-drawing optical, electrical, acoustic, and fluidic elements within a shared cross-section, thereby enabling multiple functional materials and device components to be integrated along a continuous fiber [70,71]. Additive manufacturing (AM) provides a complementary route in which geometrical and compositional features can be programmed locally rather than defined exclusively by a longitudinally uniform preform. Three-dimensional printing of optical-fiber preforms, for example, permits the fabrication of customized cross-sections, internal air channels, structured claddings, and spatially arranged material domains that can subsequently be consolidated or thermally drawn into extended fibers [72]. Hybrid approaches combining printed preforms, thermal drawing, laser processing, and selective surface deposition further enable sensing regions to be introduced at designated positions or length scales. Consequently, fiber architecture can progress from a largely uniform waveguide or conductive filament toward a hierarchical structure containing internal channels, embedded conductors, functional interfaces, and locally differentiated sensing elements [24,70–72].

Additive processes also enable sensing materials, electrodes, and interconnects to be deposited directly onto individual fibers, yarns, fabrics, and curved three-dimensional

substrates. Direct ink writing and conformal printing can form conductive traces and functional coatings along non-planar surfaces, facilitating the integration of sensing and signal-transmission components without requiring the fiber to be transferred onto a rigid planar circuit [73]. At smaller length scales, electrohydrodynamic (EHD) jet printing uses an electric field to generate fine liquid jets, allowing feature dimensions below those commonly obtained through mechanically driven extrusion [74]. Related processes, including near-field electrospinning and melt electrowriting, provide control over filament diameter, alignment, spacing, stacking sequence, and network porosity. These parameters can be used to regulate conductive percolation pathways, interfacial contact, strain localization, and the effective surface area available for mechanical, chemical, or biological interactions [74–77]. The resulting architectures may include aligned microfibers, porous fibrous membranes, patterned sensing nodes, multilayer electrode–dielectric structures, and conformal circuits distributed around the fiber circumference. Their sensing characteristics, however, depend not only on constituent materials but also on process parameters such as ink viscosity, conductivity and surface tension; applied voltage; nozzle diameter; flow rate; nozzle–substrate distance; curing conditions; layer registration; and interfacial adhesion [73–87]. Small variations in these parameters can modify filament diameter, pore morphology, conductive-network connectivity, optical loss, residual stress, and interface quality, which can in turn influence sensitivity, hysteresis, cross-sensitivity, baseline stability, and device-to-device variation. Additive manufacturing has also begun to be monitored using fiber sensors themselves; for example, chirped fiber Bragg gratings have been investigated for spatially resolved thermal measurements during material deposition, reinforces this connection between process state and measured response [88]. Thus, advanced manufacturing does more than provide an alternative method for producing fiber sensors: manufacturing history therefore becomes part of the sensor transfer function rather than a separate production detail. This development provides a physical basis for applying machine learning to structure optimization, in-process monitoring, multivariable calibration, and the interpretation of responses generated by increasingly complex fiber architectures.

1.2. The imperative for machine learning

Analytical calibration and physics-based models remain effective when boundary conditions are known, individual measurands can be isolated, and transduction parameters remain stable [31]. These assumptions become difficult to maintain in advanced fiber sensors. In wearable and tactile systems, conformal contact creates pressure and strain distributions that vary with surface curvature, friction, preload, and placement, while multimodal coupling allows several physical or environmental variables to influence the same response [89,90]. Variations in fiber diameter, material distribution, electrode geometry, and printed interconnects can further introduce device-dependent behavior, even among nominally identical sensors [2,23,78]. Repeated loading, viscoelastic relaxation, and ageing may additionally alter the sensor transfer function over time. Consequently, the relationship between the underlying physical state and the measured signal can become nonlinear, history-dependent, cross-sensitive, and non-unique, limiting the accuracy of analytical inversion and fixed calibration functions [31,32,67].

Data-driven models provide a practical way to learn these complex relationships directly from measured or simulated data. They can accelerate the optimization of sensor geometry and materials, map nonideal optical or electrical responses to physical quantities, separate coupled sensing signals, and convert the recovered information into an event, diagnosis, or control command [34,37,38,42,43,48,49,79]. Their use is therefore becoming necessary as fiber-sensing systems grow more conformal, multimodal, and

structurally complex: introduction of ML can fundamentally change what is observable from a fiber-sensor response. Rather than merely refining an existing calibration curve, a learned model can establish an inverse mapping where the relationship between the measured signal and the target variable is too nonlinear, coupled, or non-unique to be described analytically. This capability has allowed individual Bragg wavelengths to be recovered from overlapping spectra [37], the position of a perturbation to be inferred from the globally integrated speckle pattern of a multimode fiber [91], and both contact force and location to be reconstructed from the spatially coupled responses of FBGs embedded in a conformal tactile skin [7]. Spectral overlap, mode interference, and sensor crosstalk can therefore become sources of discriminative information rather than unresolved ambiguity. Where the underlying quantity is already measurable using established approaches, ML can provide a more complete or reliable interpretation of the recorded response. It has accelerated distributed Brillouin-frequency extraction while improving estimation accuracy [34], enabled rapid temperature demodulation from sparsely sampled sapphire FBG spectra under vibration [38], separated target-dependent multimode-fiber interference from strong environmental perturbations [58], and denoised distributed acoustic sensing data without requiring noise-free training targets [43]. The recovered information can subsequently support event detection, diagnosis, activity or object recognition, and closed-loop control [15,16,45,46,58,92–95]. ML thus extends fiber sensing from the measurement of an individual signal feature towards the reconstruction and interpretation of complex physical states. Nevertheless, the relevant information must first be encoded by the physical sensor, reliable intelligent sensing therefore requires the sensor architecture and the data-driven model to be designed together.

Note that these successes do not imply that a more complex model is automatically preferable. A learned mapping can exploit correlations specific to one device, participant, or laboratory setup, and it cannot recover a physical variable that the sensor response does not encode. Analytical or physics-based baselines therefore remain necessary, especially when the expected relation is low-dimensional and stable. The central question throughout this review is consequently not whether ML can fit a dataset, but which additional information it recovers, under which validation unit, and whether that gain persists beyond the device and conditions used for training.

1.3. Scope and organization of this review

This review aims to connect physical configuration, data interpretation, application models, and deployment constraints across optical and electronic fiber platforms, with a primary scope on optical, electronic, fibrous and textile-integrated sensors. Perform of end-to-end synthesis from physical design optimization through intelligent demodulation to advanced applications are organized in the following Chapters: Chapter 2 evaluates the evidence for learning- and algorithm-assisted physical design and fabrication-aware optimization. Chapter 3 examines signal conditioning, representation, demodulation, and fusion, where direct fiber evidence is more developed. Chapter 4 separates temporal and spatial inference from bidirectional interaction and feedback control, and Chapter 5 addresses data coverage, generalization, and edge deployment.

2. Toward learning-assisted physical design and spatial configuration

The physical architecture determines what information is encoded in a sensor signal prior to interpretation. Fiber geometry, material distribution, electrode configuration, network topology, and interrogation settings collectively determine how an external stimulus is converted into an optical or electrical response. Learning-assisted design can therefore target response separability [36,48–57], fabrication tolerance [75–78], and calibratability across devices and conditions [48,49] alongside conventional sensing

measures. The design problem consequently shifts from maximizing response magnitude to producing distinctions that remain recoverable by the decoder.

This design problem extends across three coupled levels. Device-level variables define the fiber cross-section, material composition, microstructure, electrode pattern, and transduction geometry. Network-level variables determine placement, routing, channel allocation, and textile connectivity, while interrogation-level variables control illumination, pulse coding, multiplexing, sampling, and excitation. A change at any level alters the data received by the model, so the physical front end and decoder are more usefully evaluated as one sensing system.

Algorithm-guided co-design has been explored through grating parameters, interrogation codes, sensor placement, and coupled sensor-decoder architectures [1,3,31,32,47,96,97]. Topology and computational reconstruction in adjacent wearable and photonic platforms broaden the available design strategies. Most evaluations, however, remain based on simulation or a single prototype rather than independently fabricated sensor instances. These studies provide useful methods for fiber-sensor development, but their performance cannot be assumed to transfer directly without fabrication and device-level validation. The following subsections therefore distinguish experimentally demonstrated fiber-sensor designs from methods that currently represent transferable design opportunities.

2.1. Geometric and topology optimization

Geometric and topology optimization offer two complementary ways to control physical encoding. Geometric optimization tunes predefined dimensions or positions, including grating parameters, cross-sectional features, electrode spacing, and sensor placement. Topology optimization instead determines how sensing materials, compliant regions, and optical or electrical pathways are distributed and connected within an allowable domain [51,52,54,57]. The distinction between refining an architecture and reorganizing it also determines which search strategy is practical. When variables are discrete, the search space is non-convex, or analytical gradients are unavailable, population-based algorithms provide a flexible search mechanism. For example, Cormier et al. coupled a real-coded genetic algorithm to an F-matrix model to recover the parameters of uniform, chirped, and apodised FBGs from prescribed spectra [51]. In a related use of evolutionary search, Sun et al. optimized an aperiodic interrogation code for improved information recovery [36], while distributed sensor placement was optimized against strain-based structural objectives [54]. The same class of algorithm can thus connect spectral, interrogation, and network objectives to different sets of design variables.

When evaluating each candidate requires an experiment, fabrication cycle, or finite-element simulation, a probabilistic surrogate can reduce the number of trials. Bayesian optimization balances predicted improvement against uncertainty and has been used to identify wearable-device parameters from noisy metabolic-cost measurements [58]. In fiber sensing, the candidate variables could include preload, grating allocation, electrode spacing, textile tension, and encapsulation geometry. Multi-objective formulations would then expose trade-offs among sensitivity, range, hysteresis, durability, channel count, comfort, and fabrication cost.

If the architecture itself is allowed to change, optimization extends from predefined dimensions to material distribution and connectivity [52,55]. Mechanical and transduction models can be coupled to a task-level objective, with learned surrogates replacing expensive components when direct differentiation is impractical. Deliberately heterogeneous responses may improve state identifiability by giving sensing elements different working ranges. Using this principle, Yang et al. applied localized topographic

patterning to full-body avatar reconstruction and achieved a 3.5 cm average position error [49]. This joint design of physical encoding and reconstruction resembles computational imaging [56,79]. Figure 2 illustrates this co-design perspective through active-learning-guided formulation (Figure 2a) [98], multimode-fiber shape encoding and task-specific perception (Figure 2b,c) [99], and neuromorphic FBG decoding (Figure 2d) [100]. Regardless of the search algorithm, the objective determines which sensing capability is favored. Spectral agreement, recovered information, strain observability, and position error have served as task-level objectives in FBG synthesis, interrogation design, sensor placement, and topographic wearables [36,49,51,54,57]. Their relevance depends on whether the improvement persists under unseen loads and realistic fabrication tolerances. When the desired response is specified before the structure is known, the same design problem takes the form of inverse design.

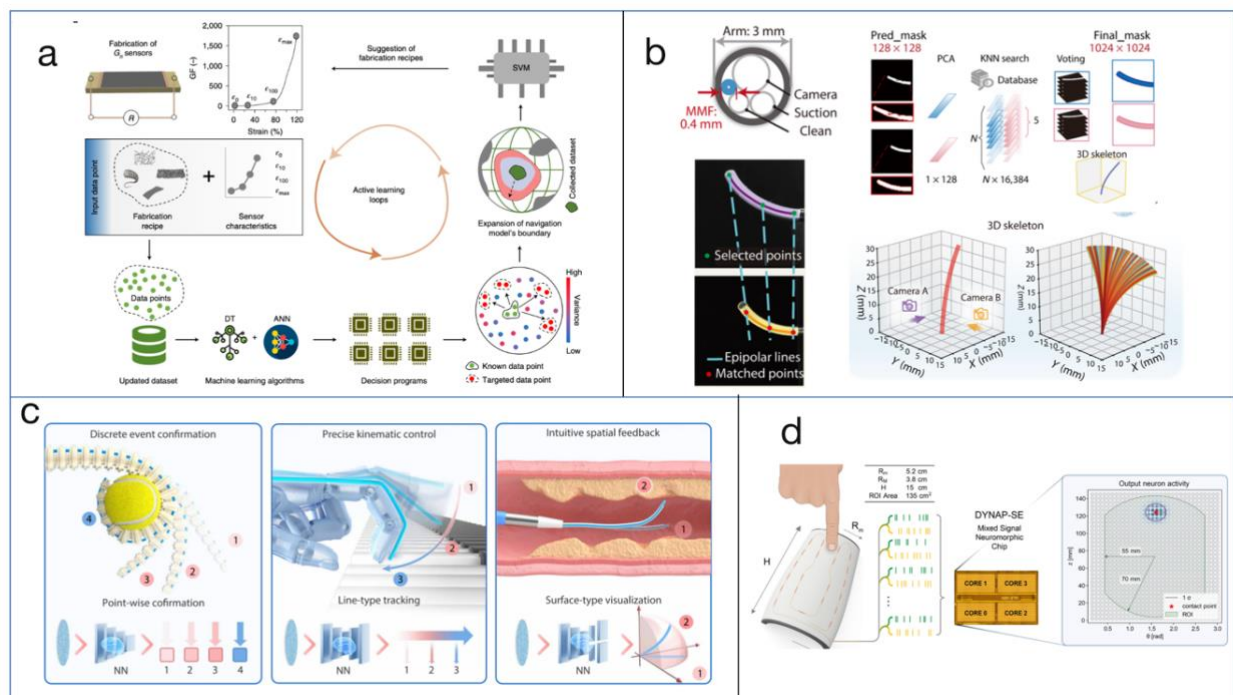


Figure 2. Representative machine-learning-enabled fiber-sensor design and perception: (a) active-learning-guided strain-sensor design that iteratively selects fabrication recipes from sparse training data (Adapted and reproduced with permission from Ref. [98]. Copyright 2022, Springer Nature); (b) single-ended multimode-fiber shape reconstruction using segmentation, principal-component analysis, nearest-neighbor matching, and stereo geometry (Adapted from Ref. [99]. Copyright 2026, American Association for the Advancement of Science; licensed under CC BY 4.0); (c) a deep-learning-enabled multimode-fiber platform applied to discrete-event confirmation, kinematic tracking, and spatial-feedback tasks (Adapted from Ref. [99]. Copyright 2026, American Association for the Advancement of Science; licensed under CC BY 4.0); and (d) an FBG tactile skin coupled to a mixed-signal neuromorphic processor for contact localization (Adapted from Ref. [100]. Copyright 2026, Springer Nature; licensed under CC BY 4.0).

2.2. Machine learning for inverse design

Inverse design begins with a desired optical, electrical, or mechanical response and searches for a structure that produces it. Neural surrogates reduce repeated geometry-to-spectrum simulation in nanophotonics, where expensive electromagnetic models and non-unique solutions impede iterative search [48,50,55]. For example, Tahersima et al. trained a topology-to-transmission model that generated integrated-photonics power splitters for prescribed responses without rerunning a full simulation at every iteration

[50]. Fiber implementation introduces additional constraints from cylindrical or textile coordinates, bendability, multimaterial interfaces, connectorization, feature size, and fabrication tolerance. Relevant targets include grating profiles, photonic-crystal-fiber cross-sections, coatings, and multimaterial distributions.

Three learning strategies address different parts of this inverse problem. Tandem networks evaluate a generated design with a pretrained forward model when several structures can produce similar responses [48,55]. Latent-space optimization restricts the search to a manufacturable design family before Bayesian or gradient-based refinement. Active learning instead selects the simulations or experiments with the greatest model uncertainty. In an adjacent strain-sensor platform, Yang et al. seeded an SVM with 351 nanocomposite formulations, completed 12 active-learning cycles with 125 additional fabricated sensors, and then applied genetic optimization across more than 10,000 virtual samples [98]. Uncertainty-guided sampling reduced the experimental search burden in that setting, although its transfer to fiber fabrication remains untested. For optical fibers, the target response may emphasize spectral separation, mode sensitivity, spatial resolution, confinement loss, or rejection of temperature and strain cross-sensitivity. Finite-element modelling has related the cross-section and Au-TiO₂ interface of a photonic-crystal-fiber surface-plasmon-resonance sensor to its refractive-index response [57]. Although no learned inverse model was used, the resulting geometry–response map could supply the forward data needed by a surrogate. Candidate structures would then be proposed on the learned response surface and verified with independent multi-physics calculations.

Table 1. Representative evidence for physical design and fabrication-aware optimization.

Objective	Design variable	Method	Reported validation	Evidence boundary	Ref.
Synthesize an FBG ¹ with a specified reflection spectrum	Grating parameters	Real-coded genetic algorithm	Synthesis of a target Bragg-grating spectral response	Direct fiber design; numerical synthesis	[51]
Improve information recovery through interrogation-code design	Aperiodic interrogation code	Genetic optimization	Experimental coding gain with Brillouin and Raman sensing	Direct optical-system validation	[36]
Optimize sensing coverage and fiber routing for structural monitoring	Sensor position and coverage	Genetic algorithm	Placement optimized for structural sensing objectives	Structure-specific network topology	[54]
Match sensor working ranges to body-joint deformation and support motion reconstruction	Surface topography and working window	Pattern design plus in-sensor ML ²	Full-body classification and avatar reconstruction; 3.5 cm average error	Direct wearable prototype and edge chip	[49]
Relate photonic-crystal-fiber geometry to refractive-index response	Cross-section and Au-TiO ₂ interface	FEM ³ with AI-assisted ⁴ optimization	Predicted sensitivity and geometry-response trade-offs	Fiber-specific numerical evidence; no fabricated comparison	[57]
Recover subsurface thermal fields during additive manufacturing	Sensing and process-data configuration	Learned high-resolution temperature reconstruction	Sub-surface thermal measurement during fabrication	Direct fiber measurement; adjacent to sensor-fabrication control	[88]

¹ FBG: fiber Bragg grating. ² ML: machine learning. ³ FEM: finite element method. ⁴ AI: artificial intelligence.

Lastly, the selected structure also should remain feasible during manufacture. Conductive-path continuity, curvature, minimum feature size, stress concentration,

optical loss, connector count, washability, and wearer comfort can be represented as 328
constraints or secondary objectives [78]. Existing demonstrations still emphasize learned 329
or simulated forward models in photonic and PCF systems [48,50,55], leaving fabrication- 330
aware experimental validation comparatively limited. Simulation can therefore narrow 331
the design space, but fabrication is still needed to reveal the structured deviations 332
introduced during production. 333

2.3. Mitigating fabrication imperfections via data perspectives

Designs constrained by manufacturing processes may still exhibit structural variations during production. For printed and fibrous devices, structured variation can arise from ink rheology, flow, nozzle motion, curing, adhesion, electric field, fiber diameter, alignment, porosity, yarn tension, loop geometry, and junction resistance [75,78]. ML-based additive-manufacturing studies use acoustic, thermal, image, and point-cloud data to detect abnormal states or predict surface quality. These methods suggest useful monitoring features, but they do not establish that a fiber sensor fabricated under variable conditions will retain calibration. For example, Wang et al. provide a complementary direct fiber measurement: a chirped FBG and neural network reconstructed subsurface temperature during laser powder-bed fusion [90]. The system measures an additive-manufacturing process; it does not close the loop on fabrication of the sensing fiber itself. Related manufacturing routes embed digital sensing and data handling during thermal drawing (Figure 3) [101]. In a separate process, Mieszczanek et al. stabilized melt electrowriting through neural feedback [102]. Together, these elements connect process monitoring with subsequent model calibration, even though a complete closed loop for fiber-sensor fabrication has yet to be tested.

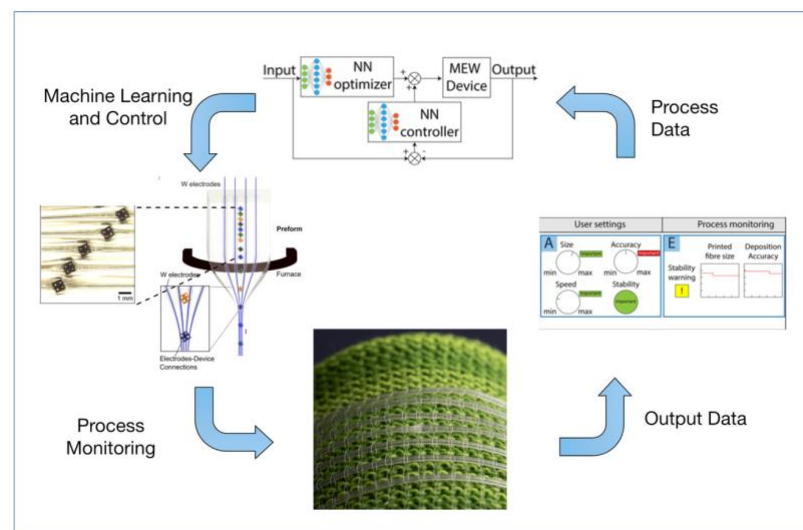


Figure 3. Closed-loop fabrication and process control for intelligent fibers: digital devices are integrated during thermal drawing and embedded in textiles for sensing and in-fiber inference (Adapted from Ref. [101]. Copyright 2021, Springer Nature Limited; licensed under CC BY 4.0), while neural-network optimization and feedback control use real-time process data to stabilize melt electrowriting and monitor output quality (Adapted and reproduced with permission from Ref. [102]. Copyright 2024, Springer Nature).

The same process measurements can be carried back into design and model training. Distributions of fiber diameter, contact resistance, adhesion, and printing gaps can be introduced into simulation to estimate how manufacturing variation changes sensor performance. Training the decoder on these responses may improve stability across

realistic device variation. Such process-informed perturbations are more representative than independent Gaussian noise because fabrication errors are often spatially correlated and dependent on material state or loading [76,77]. They also help distinguish variations that calibration may correct from open circuits, delamination, or grating fracture that require fault detection or replacement. Linking these records through a device-level identifier would complete the data pathway from manufacture to sensing performance. Process-monitoring records can be associated with structural characterization, calibration results, and final response [80–86]. Across several batches, this digital thread could reveal which deviations remain compensable and which predict irreversible failure. It would also provide a traceable basis for testing whether calibration transfers between independently manufactured sensors.

3. Signal processing and intelligent demodulation pipelines for fiber sensors

Once the physical sensor has generated a response, the next task is to convert that response into a stable physical estimate. The required processing depends on whether acquisition noise [33,34], material memory [2,23], or cross-sensitivity [39] perturbs the signal or changes its relationship to the measurand. The signal pathway provides a natural organization for this problem. Noise, drift, and hysteresis are first characterized so that weak responses are not removed with unwanted variation. The retained temporal, spectral, and spatial structure is then represented through engineered or learned features. Finally, coupled responses are deciphered or combined with complementary modalities to recover physical states. Method choice throughout this pathway depends on physical consistency, data and calibration coverage, computational cost, and transfer across devices and operating conditions.

3.1. Noise reduction and drift compensation

Noise enters sensor data through the acquisition chain and data collecting environment. During data acquisition phase, shot noise, thermal noise, electronic readout noise, laser fluctuations, digitization effects and finite spectral resolution may arise [33,38]. Distributed optical measurements add phase fading, polarization variation, Rayleigh-speckle fluctuations, attenuation, and channel-dependent gain [11,13,65,103–106]. Electrically functional fibers and wearables are further affected by contact resistance, viscoelastic creep, motion artefacts, temperature, humidity, preload, mounting, and repeated donning [70]. Locating the dominant source narrows the set of plausible correction methods. These sources differ in both time scale and statistical structure. Shot and readout noise are commonly broadband and weakly correlated, whereas laser and optical-power fluctuations may appear coherently across channels. Impulse artefacts are sparse, baseline drift develops slowly, and hysteresis produces different loading and unloading responses at the same instantaneous input. Persistent sensitivity loss or broken channels instead indicates degradation. Power spectra, autocorrelation, cross-channel coherence, environmental residuals, and loading–unloading loops help separate these mechanisms, while repeated cycles, sessions, and devices distinguish temporary variation from structural change.

Given the diverse characteristics of the signals, appropriate correction methods can be selected based on the dominant mechanism. When noise and signals are distinguishable in terms of frequency or scale, filtering techniques such as band-pass filtering, median filtering, wavelet filtering, empirical mode decomposition (EMD), and variational mode decomposition (VMD) prove effective [33,35,107]. Reference channels and common-mode subtraction can be employed to address coherent fluctuations, while

measured interference variables facilitate multivariate compensation [39,63–65]. State-space tracking is suitable for handling slowly varying baselines, and sequence models can capture historical dependencies; meanwhile, learning-based denoising methods offer advantages when noise and signal frequency bands overlap. As shown in Figure 4a, Lalam et al. improved the signal-to-noise ratio of two Brillouin peaks by approximately 4 dB and reduced the measurement uncertainty of one of the frequency shifts from 3 MHz to 1.2 MHz [108]. However, the extent of correction is constrained by the requirement to preserve the characteristics of weak signals.

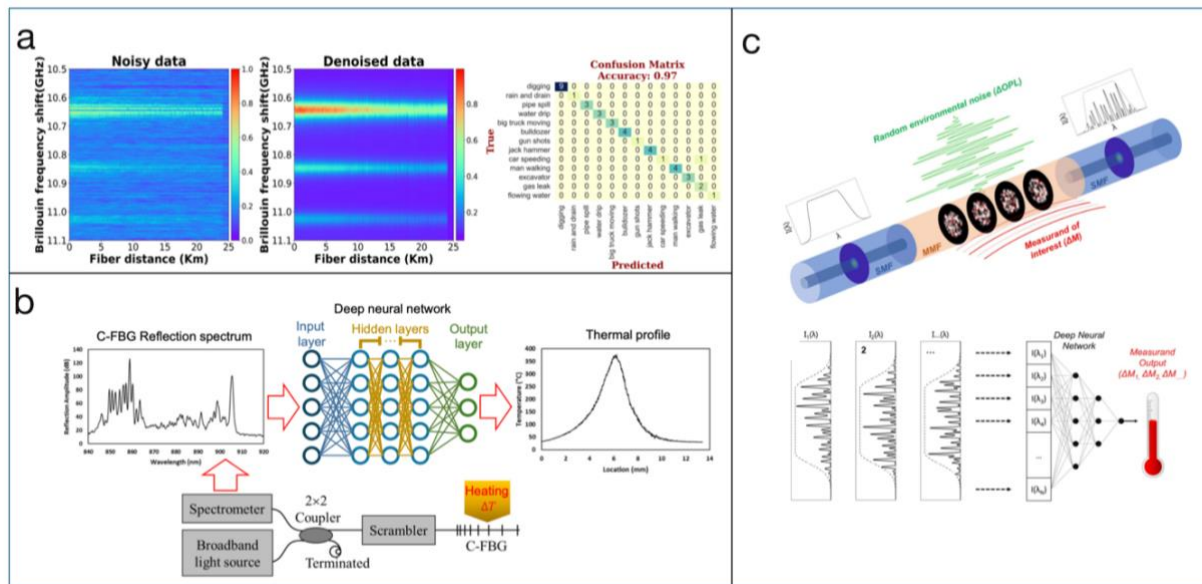


Figure 4. Learning-based denoising and robust demodulation in fiber sensors: (a) a deep-learning workflow denoises distributed Brillouin frequency-shift maps and classifies pipeline events (Adapted and reproduced with permission from Ref. [108]. Copyright 2024, Springer Nature); (b) a deep neural network maps chirped-FBG reflection spectra to subsurface thermal profiles during additive manufacturing (Adapted from Ref. [88]. Copyright 2024, Springer Nature; licensed under CC BY 4.0); and (c) a multimode-fiber interference sensor uses a deep neural network to recover the target measurand in the presence of strong optical-path-length noise (Adapted and reproduced with permission from Ref. [58]. Copyright 2021, Chinese Laser Press).

When clean reference signals are unavailable, the training signal can instead come from consistency across repeated or correlated measurements. Building on paired DAS fibers with independent noise, Lapins et al. trained DAS-N2N to enhance coherent weak events without clean labels and process a 30 s, 985-channel record in less than 1 s [43]. Blind DAS denoising extends the same principle to measurements without paired fibers [59]. For spectral sensing, deep denoising has also improved experimentally acquired responses from an FBG glucose sensor [60]. These gains are most informative when accompanied by amplitude bias, spectral fidelity, waveform distortion, and rare-event preservation rather than signal-to-noise ratio alone.

A different problem arises when environment, sensor history, or spectral interference changes the signal-to-measurand relationship itself. Calibration-aware models can incorporate temperature, humidity, preload, and other conditions as measured inputs or latent states. Figures 4b and 4c illustrate related deep models that reconstruct thermal fields from chirped-FBG spectra [88] and recover target measurands under optical-path-length noise [58], respectively. Overlapping FBG spectra have been demodulated through

shared LSTM extraction [37] and generative spectral reconstruction followed by dense regression [61]. Coupled strain and temperature have likewise been separated using multiple FBG arrays [39]. For viscoelastic sensors, training data also need loading and unloading, minor loops, variable rates, dwell periods, and recovery so that sequence or state-space models can represent path dependence. A static monotonic calibration cannot resolve the same instantaneous signal when it occurs on different loading histories.

Correction is therefore incomplete without an indication of reliability. Varying each nuisance source and reporting downstream error, weak-event recall, waveform or spectral fidelity, and calibration stability can reveal when a model has left its validated range. Open-set vibration recognition using a residual CNN and OpenMax has identified previously unseen event classes in a distributed-fiber system [106]. The result supports abstention for unfamiliar events, although it does not imply coverage of every hardware fault. Similar confidence signals could preserve traceable pre- and post-update errors during online calibration.

3.2. Feature engineering and representation learning

Feature engineering converts raw sensor data into compact descriptors that retain information relevant to the sensing task. The appropriate descriptors depend on the signal format, transduction mechanism, and target measurand. Time-domain features commonly include the mean, variance, slope, peak count, rise time, and relaxation time [109–111]. History-dependent responses can be described by the loading–unloading area, residual offset, or recovery rate. Optical spectra can be represented by the Bragg wavelength, bandwidth, spectral centroid, peak ratios, and harmonic energy [37,38]. Short-time Fourier and wavelet coefficients capture changes in frequency content over time [15,107]. Distributed traces can also be described by event velocity, inter-channel delay, spatial gradients, and the extent of the activated sensing region [15,95]. Feature selection should therefore be guided by the transduction physics and the intended output rather than by a generic collection of statistical descriptors. Once candidate descriptors have been defined, redundant variables can be removed through correlation analysis, mutual information, principal-component analysis, recursive feature elimination, or model-based importance. The retained vector is normalized before classification or regression. These steps keep the model input compact and expose which descriptors drive the prediction.

The resulting feature vectors are compatible with support-vector machines, random forests, gradient-boosted trees, and shallow neural networks [15,95]. These models require comparatively little training data, remain interpretable, and can run on constrained hardware. For example, Bublin reported accuracies above 99% for several feature-based classifiers in DAS excavator detection, whereas a CNN achieved similar accuracy with sixfold lower detection delay and twelvefold lower execution time [95]. Engineered features are therefore attractive within a stable operating regime, but their maintenance becomes more difficult as event diversity and environmental variation increase.

When informative structure cannot be expressed as a stable feature vector, representation learning shifts descriptor construction into the model. One-dimensional CNNs capture local temporal or spectral motifs, LSTMs encode history, transformers connect distant observations, and two-dimensional CNNs exploit spatial organization. By transferring representations from seismometer waveforms, Hernández et al. achieved 96.94% earthquake-detection accuracy on fiber-optic DAS [16]. A U-Net reconstructed images from multimode-fiber temporal waveforms with 81.8% average fidelity, a structural-similarity index of 0.78, and 15.4 million frames per second [112]. Task-specific decoders applied to single-ended speckle fields further exceeded 99% discrete-state

accuracy, improved continuous-tracking resolution about fivefold, and reached an intersection-over-union above 0.93 for soft-robot reconstruction [99]. These outcomes reflect the value of retaining structure that manual summaries may discard.

Between fixed descriptors and fully end-to-end learning, hybrid pipelines retain known physics while learning residual structure. Gain correction, alignment, and restriction to meaningful spectral or frequency bands can precede learned encoding. When labels are scarce, paired-noise, contrastive, or masked-reconstruction objectives provide temporal and spectral context from unlabeled measurements [43,59]. The augmentation remains valid only if it preserves the target quantity. Amplitude scaling may aid event classification, for example, while biasing force or temperature regression. The appropriate balance depends on data coverage and deployment conditions. Engineered features often remain competitive with few independent devices, stable physical descriptors, or strict interpretability and power requirements [95,109–111]. End-to-end models may be more useful when larger datasets contain overlapping temporal, spectral, or spatial patterns [16,112]. Hybrid and self-supervised approaches occupy the intermediate case, where unlabeled or physics-guided structure is available but manual labels are limited [43,59,95].

Whichever representation is selected, the validation split determines what form of transfer has been tested. Subject-, session-, device-, batch-, and site-wise partitions expose different sources of variation and are more informative than random division of adjacent windows. Feature selection and augmentation remain inside each training fold to avoid leakage. Physics-aware baselines and ablations then separate the contributions of preprocessing, augmentation, and architecture, while physical interventions test whether the learned representation uses the expected signal regions.

3.3. Multi-modal data fusion mechanics

Multimodal inference in fiber sensing includes two technically distinct problems. In cross-sensitive fibers, a shared response is decoded into several measurands such as strain, temperature, humidity, pressure, or biochemical concentration. In multisensor systems, fiber channels are combined with inertial, optical, acoustic, physiological, or visual observations. Demixing and sensor fusion therefore differ in their supervision, identifiability assumptions, and validation procedures, even when both use multimodal learning architectures. For multi-sensor systems, the fusion architecture depends on where the channels are combined. Early fusion concatenates synchronized inputs but is sensitive to scaling, missing data, and sampling mismatch. Intermediate fusion combines modality-specific latent features through attention, gating, or tensor operations, whereas late fusion combines decisions and better tolerates missing channels. Hybrid designs can use several levels at once. For example, Mao et al. combined tactile and visual information for dexterous robotic housekeeping, while other wearable systems illustrate related fusion strategies [46,113–116]. For fiber sensing, a shared latent state can combine the fiber response with contextual measurements such as temperature or inertial signals.

Table 2. Representative evidence of signal processing and intelligent demodulation methods.

Objective	Data source	Method	Output performance	Validation	Ref.
Brillouin frequency-shift extraction	Two-dimensional Brillouin spectra with low SNR	Fully convolutional network	Higher extraction accuracy and shorter processing than Lorentzian curve fitting	Simulation plus one experimental platform	[34]
Wavelength demodulation in multiplexed FBG networks	Overlapping spectra	Model-sharing LSTM	High-accuracy, high-speed wavelength extraction under	Configured network; cross-	[37]

Sparse sapphire-FBG thermometry	Coarsely sampled, irregular spectra	Machine-learning regressor	overlap; model reused without retraining 0.23 °C accuracy at 5 kHz; approximately tenfold accuracy gains over peak or curve fitting	device transfer not central Sensor- and setup-specific calibration [38]
Self-supervised DAS denoising	Paired fibers with independent random noise	Noise2Noise network	30 s × 985 channels processed in <1 s with weak-event enhancement	Field data; paired-noise assumption [43]
Multimode-fiber decoding under strong noise	Dynamic modal interference and strong noise	Deep learned inverse	Recovery of sensing information inaccessible to standard signal processing	Proof of concept on one optical system [58]
Plasmonic biosensor demodulation	Complex comb-like spectra	Spectral regression	Surface refractive-index dynamics recovered from the full spectrum	Fabricated sensor; chemistry and device specific [44]
Strain–temperature demixing	Coupled responses from multi-array FBGs	Machine-learning demixing	Discrimination of strain and temperature variations	Direct structural-sensing study; batch transfer unclear [39]

¹ SNR: signal to noise ratio. ² FBG: fiber Bragg grating. ³ LSTM: machine learning. ⁴ DAS: distributed acoustic sensing.

Regardless of fusion level, the channels need a defensible temporal and spatial alignment. Optical and textile signals may differ in latency, clock, and effective bandwidth, while propagation through a garment introduces state-dependent delay. Experimental synchronization, cross-correlation, learned offsets, or asynchronous attention can reduce small mismatches. Simulated channel loss during training can also prevent a failed yarn or missing camera from collapsing the complete system. Even well-aligned channels may not contribute equally. A high-quality camera or laboratory force plate can dominate a noisier fiber response during training, leaving the deployed model dependent on a modality that will later be absent. Auxiliary sensors used only for labels are therefore separated from channels available in operation. Modality dropout, bottlenecked fusion, and teacher–student training can preserve information in the deployed fiber pathway, and performance can be reported for both the full system and every deployable subset.

Modality ablation provides a direct test of whether fusion adds task-relevant information. For instance, Liu et al. classified 11 objects with a tactile–olfactory system at 96.9% accuracy, compared with 81.9% for touch and 66.7% for smell alone (Figure 5a) [45]. Controlled corruption and cross-user or cross-device evaluation can then determine whether the advantage persists beyond the training conditions. Attention weights are less conclusive because they may encode wearer or device identity rather than the target state.

4. Advanced learning paradigm across sensing applications

Following the demodulation of signals, the learning task shifts from physical quantity estimation to application-specific inference. Human-motion monitoring typically benefits from time-series models capable of parsing movement phases, state transitions, or continuous trajectories; tactile skins and distributed arrays rely more on spatial models that interpret the distribution of mechanically coupled fields; and interactive and robotic systems—where estimation results directly inform feedback or control—face constraints regarding latency and stability. These temporal, spatial, and closed-loop application

scenarios each have distinct priorities, thereby dictating the appropriate model architectures and evaluation criteria.

4.1. Time-series models for dynamic human motion

Dynamic human-motion sensing is the continuous measurement and interpretation of body movement from time-varying sensor responses [109,117]. Unlike static posture detection, it describes how movement develops over time. Depending on the application, the target may be an activity label, a sequence of movement phases, or a continuous variable such as joint angle, limb trajectory, gait symmetry, or movement speed [118–120]. Figure 6 illustrates this range through avatar reconstruction (a) [49], sign-to-speech translation (b) [121], neurological screening (c) [122], sign-language translation (d) [6], and continuous finger-motion regression (e) [123]. Similar instantaneous sensor values can arise from different movements or from different phases of the same movement, making isolated samples ambiguous [124,125]. To capture this temporal information, fiber and textile systems record movement as synchronized signal sequences from one or more sensing channels. Optical-fiber wearables may encode deformation through intensity, wavelength, or bending-loss changes [5,126], whereas functional yarns and textile arrays commonly produce resistance or pressure variations [6,127–131]. These signals can support several forms of inference, including activity classification, temporal segmentation, event detection, and continuous regression. Importantly, these outputs should be distinguished because recognizing a completed gesture does not establish that a model can reconstruct its trajectory or identify its onset and completion [109,119,120].

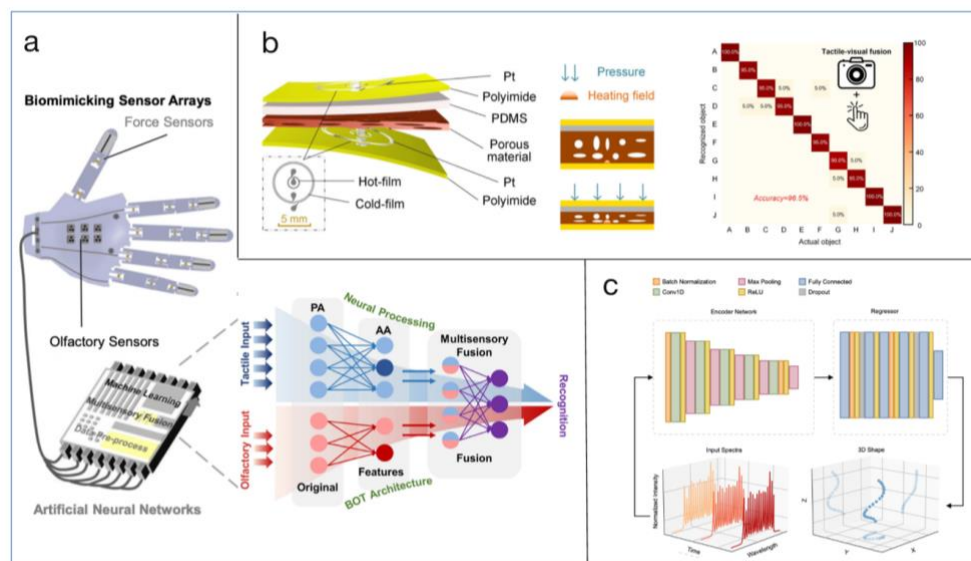


Figure 5. Learned spatial representations and multimodal fusion: (a) neural fusion of tactile and olfactory signals for robust object recognition (Adapted from Ref. [45]. Copyright 2022, Springer Nature; licensed under CC BY 4.0); (b) a pressure–temperature tactile element and tactile–visual fusion pipeline for object classification (Adapted from Ref. [46]. Copyright 2024, Springer Nature; licensed under CC BY 4.0); and (c) a one-dimensional convolutional encoder–regressor that reconstructs three-dimensional catheter shape from multicore-fiber spectra (Adapted from Ref. [9]. Copyright 2023, MDPI; licensed under CC BY 4.0).

Resolving this ambiguity requires representations that preserve phase, duration, and ordering. Windowed statistics and time–frequency coefficients can describe short, repeatable gestures [109–111], while cross-channel phase and correlation capture coordination between body regions [119,120]. Dynamic time warping and hidden Markov models accommodate variation in movement speed and transitions between predefined

states [109,117]. Treating each window as an independent static sample discards these relationships. Compact feature sets can remain effective when activities are discrete and sensing conditions are controlled. For example, Xu et al. evaluated a 200-point textile smart sleeve on 18 activities from 14 participants using SVM and CNN models [130]. Ten-fold cross-validation produced accuracies of 82.02% and 82.30%, respectively. Under leave-one-subject-out evaluation, the corresponding values fell to 72.66% and 74.84%. The larger change across validation protocols than across model types suggests that cross-subject transfer can dominate the apparent benefit of model complexity.

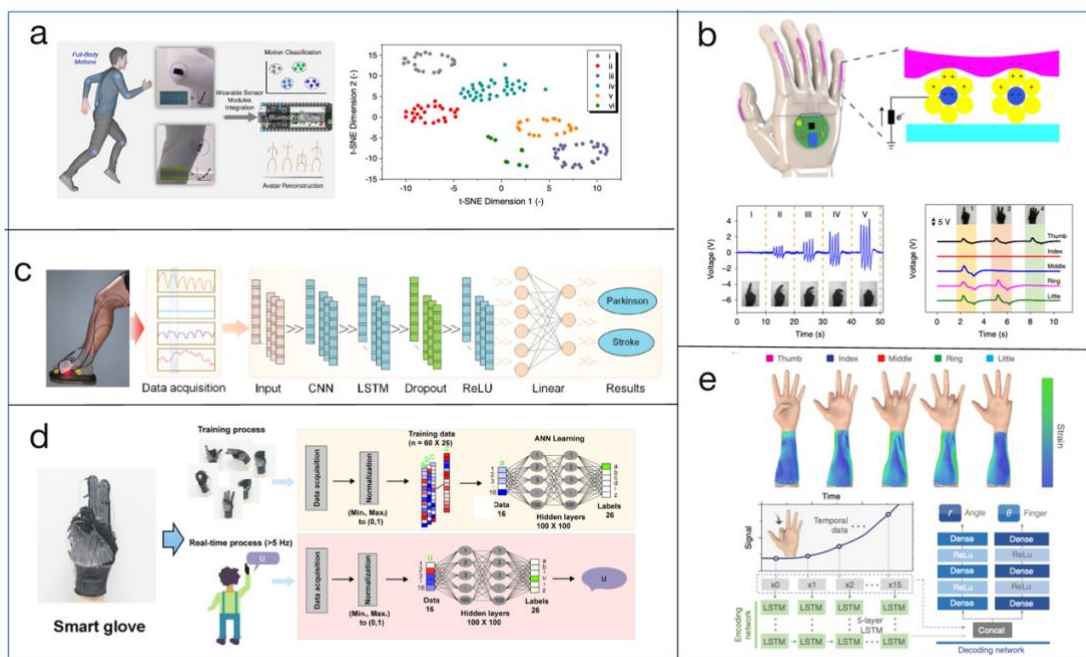


Figure 6. Temporal inference and wearable architectures for dynamic human-motion sensing: (a) distributed wearable sensor modules with edge learning for motion classification and full-body avatar reconstruction (Adapted from Ref. [49]. Copyright 2022, Springer Nature; licensed under CC BY 4.0); (b) a yarn-based stretchable sensor array and wireless electronics for machine-learning-assisted sign-to-speech translation (Adapted and reproduced with permission from Ref. [121]. Copyright 2020, Springer Nature); (c) a self-powered dual-mode insole combining triboelectric and stretchable optical-fiber signals with CNN-LSTM inference for Parkinson's disease and stroke screening (Adapted from Ref. [122]. Copyright 2026, Wiley-VCH GmbH, Weinheim; licensed under CC BY 4.0); (d) a full-fiber auxetic-interlaced smart glove and artificial neural network for real-time sign-language translation (Adapted from Ref. [6]. Copyright 2022, Springer Nature; licensed under CC BY 4.0); and (e) a skin-like strain sensor with recurrent encoding and regression of finger and joint motion (Adapted from Ref. [123]. Copyright 2020, Springer Nature; licensed under CC BY 4.0).

When motion phase or sensor history influences the output, models should operate on less processed sequences. CNNs capture local waveform motifs, LSTMs retain loading and movement history [124,125], and CNN-LSTM models combine the two functions [113,132]. Transformers can connect distant parts of long or variable activities, although their data and computation requirements are higher [133–135]. The distinction between discrete and continuous motion is reflected in the models used by fiber and textile systems. For example, Li et al. combined five D-shaped plastic-optical-fiber sensors with an SVM and achieved 99.8% accuracy for 13 gestures and 97.7% for 11 grasping actions [5]. In a related glove, 16 full-fiber auxetic sensors and an artificial neural network recognized the 26 English letters with 99.8% accuracy and produced a translation within 0.25 s [6]. These

compact classifiers suit short, separated movements, whereas phase-resolved or continuous outputs require temporal memory. The metric set depends on the intended inferential target. Completed-gesture studies commonly report classification accuracy, whereas biomechanical time-series and clinical gait studies also examine trajectory error, event timing, latency, subject-independent performance, and longitudinal stability [120,136–138]. These additional measures reveal errors that a single completed-gesture label unable to capture.

Beyond completed activity labels, temporal models can support diagnosis and continuous-state reconstruction. A CNN–LSTM fused plantar pressure with three optical-fiber strain channels for neurological screening and reached 94.23% subject-wise accuracy across 60 participants, compared with 40.38% and 88.46% for the individual modalities [122]. In a separate wrist-mounted system, an LSTM encoder and dense regressors decoded continuous finger motion [123]. Transfer learning reduced calibration for a new user or sensor position to 8 s of data per finger and about 5 min of retraining. The first result reflects complementary sensing, while the second addresses recalibration after user or placement change.

4.2. Spatial-temporal networks for electronic skins and arrays

Spatially distributed fiber and textile sensors differ fundamentally from collections of independent point sensors. An applied load is redistributed through yarn contacts, compliant substrates, encapsulation layers, and conformal interfaces before it reaches the individual sensing elements. The measured response is therefore a coupled spatial field whose structure reflects both the external stimulus and the mechanics of the sensor network. Neighboring channels may respond to the same event with different amplitudes and delays; stretching can change the relative positions of sensing nodes; and local defects can remove or distort parts of the field. The resulting learning problem is not simply channel-wise classification, but inverse spatial inference from a deformable, nonuniform, and potentially incomplete measurement lattice.

Measurement geometry can guide architecture selection. Two-dimensional CNNs have been used for regularly addressed tactile arrays, where shared kernels capture local contact patterns [139]. A temporal dimension can be added through three-dimensional convolution or a sequence model when contact onset, sliding, and release is relevant. For irregular or sparsely activated layouts, graph models represent sensing elements as nodes and mechanical or electrical relationships as edges [140,141]. Message passing then aggregates information along the physical network without imposing a fixed image grid.

For regularly addressed arrays, overlapping responses can be treated as a spatial image rather than independent channels. For example, Massari et al. combined an FBG soft skin with convolution and multigrid integration, achieving median contact errors of 35 mN and 3.2 mm [7]. On a regular tactile array, a CNN improved object recognition by as much as 17.72 percentage points over a SURF–SVM pipeline [139]. Both results rely on shared local structure, which becomes harder to preserve when the sensor layout is irregular. If the grid assumption fail, graph models can preserve physical or functional relationships between sensing nodes. By converting tactile measurements into graphs, Fan et al. reached 99.53% maximum object-recognition accuracy [141]. TacGNN extended graph-based perception to in-hand manipulation, reducing position-prediction error to 0.096 cm and transferring the tactile control policy to a physical robot without additional training [140]. These electronic-skin and robotic results do not establish equivalent performance for fiber sensors, but their graph construction is directly relevant to irregular fiber and textile networks.

At full-hand scale, the same spatial principle can be shared across a much larger array. Figure 7b shows how a knitted tactile glove with 548 sensing elements used a

shared-weight convolutional network to learn grasp signatures for 26 objects and estimate object weight [142]. A related dual-branch CNN for simultaneous material and texture recognition is illustrated in Figure 7c [143]. This electrically addressed textile example complements localized and graph-based sensing by showing how repeated spatial features can be reused across the hand.

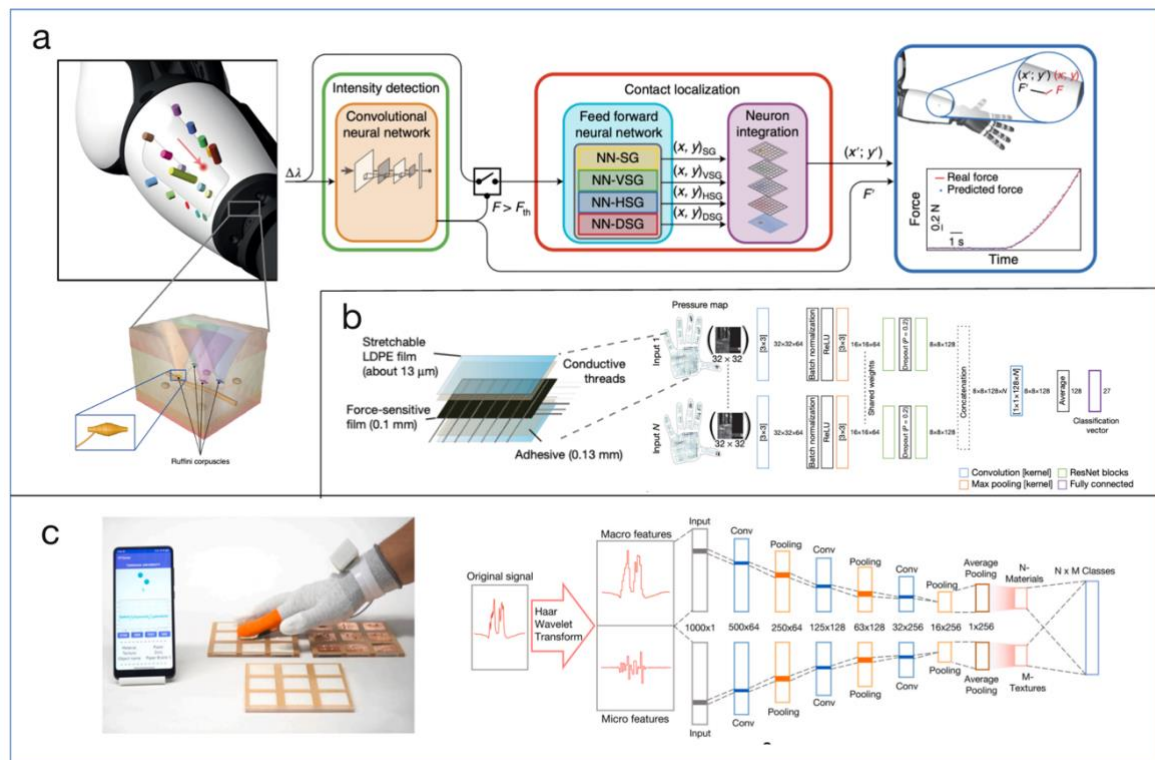


Figure 7. Spatial learning for tactile skins and sensor arrays: (a) a distributed-FBG skin with convolutional and multigrid neural networks for contact localization and force estimation (Adapted from Ref. [7]. Copyright 2022, Springer Nature; licensed under CC BY 4.0); (b) a scalable tactile glove that maps pressure images from knitted sensor arrays and classifies grasped objects with a shared-weight residual network (Adapted and reproduced with permission from Ref. [142]. Copyright 2019, Springer Nature); and (c) a triboelectric tactile glove that separates wavelet-derived macro- and micro-features with a dual-branch CNN for simultaneous material and texture recognition (Adapted and reproduced with permission from Ref. [143]. Copyright 2021, Elsevier).

Across grid and graph architectures, mechanical coupling separates nominal taxel pitch from effective spatial resolution. Each channel measures a blurred neighborhood, and overlapping sensitivity fields can make reconstruction non-unique. Apparent sub-pitch resolution may therefore reflect object-specific priors rather than recovery of the true contact field. Unseen shapes, continuously varied positions, loading trajectories, and independent force references provide stronger tests of spatial reconstruction. Non-negative pressure, force and moment balance, coherent contact patches, and friction-consistent motion can further constrain solutions that fit labels but violate contact mechanics. The network structure can also support local encoding rather than purely centralized reconstruction. Electronic-skin and tactile systems combine transduction with adaptation or learned feature extraction [27,28,140,144–148], while an FBG tactile skin coupled to a mixed-signal neuromorphic processor provides a fiber-relevant example [100]. Event-driven representations may reduce data traffic when contacts change sparsely, although the benefit depends on acquisition and hardware. Because adaptation

changes the effective response, reversible sensory state still has to be distinguished from drift before local plasticity can be treated as stable computation.

4.3. Closed-loop systems and human-machine interactions (HMIs)

Once a learning model can estimate motion, contact, force, or shape, the next stage is to use that estimate to guide feedback or action. This transition distinguishes a sensing demonstration from a closed-loop system. In an open-loop application, the model produces a label or physical estimate for later interpretation. As summarized in Table 3, a closed-loop estimate influences a controller, robotic actuator, user interface, rehabilitation cue, or operational decision. Prediction latency, temporal stability, uncertainty, and failure handling therefore become as important as average accuracy. Representative closed-loop and application-level systems are summarized in Table 3.

Soft robotics makes the coupling between sensing and action explicit. Compliant actuators and embedded sensors have nonlinear, time-varying responses that complicate analytical state estimation. For example, Thuruthel et al. used recurrent networks to infer both kinematics and contact forces from redundant embedded sensors [29]. In a subsequent feedback system, a stateful LSTM compensated for drift and hysteresis and controlled an anthropomorphic finger at 25 Hz with a 0.17 N average steady-state error [71]. Temporal learning therefore supplied a usable control variable within the tested dynamic range.

The same closed-loop logic extends to wearable interfaces, where body motion becomes a command, text output, or virtual representation. A stretchable optical-fiber sensor captured facial motion, gait, respiration, finger bending, and gestures for virtual-reality interaction and remote robotic control [30]. Bidirectional smart gloves combine this sensing pathway with communication and feedback. One triboelectric system recognized words and sentences with 91.3% and 95% accuracy, respectively (Figure 8a) [21]. Although the transduction was not fiber based, the architecture shows how sensing, inference, communication, and user feedback can operate together. Once the interface returns information to the user, feedback becomes part of the sensing pathway. In a related HMI, Sun et al. combined tactile and temperature sensing with vibrotactile and thermal feedback in a wearable ring, while AI supported gesture and object recognition [149]. A comparable fiber-based system could pair compliant sensors with textile actuators, optical indicators, or haptic elements. The two directions are evaluated together because feedback changes user behavior and may alter the input distribution received by the model.

Clinical deployment introduces a distinction between monitoring and intervention. Continuous blood-pressure monitoring has combined optical-fiber sensing, machine learning, wireless communication, and a smartwatch interface [8]. A multicore-fiber catheter reconstructed with a particle-swarm-optimized neural network achieved average distance errors of 0.57 mm under constant temperature and 0.36 mm under variable temperature [9]. These results support monitoring and navigation functions within the tested settings. Incorporation into treatment or feedback would require additional evaluation across patients, sensor instances, placement conditions, and clinically relevant failure cases.

Table 3. Representative application and deployment studies.

Application	HMI output	ML model	Performance	Key features	Ref.
Plastic-optical-fiber glove	Gesture and grasp labels	SVM ¹ and feedforward neural network	99.8% for 13 gestures and 97.7% for 11 grasps	Five-channel D-shaped POF ² sensing glove	[5]

Full-fiber sign-language glove	Translation of 26 letters into text	Artificial neural network	99.8% accuracy with a 0.25 s translation time	Sixteen auxetic-interlaced yarn sensors [6]
Textile smart sleeve	Recognition of 18 daily activities	SVM and CNN	82.02% and 82.30% in 10-fold validation; 72.66% and 74.84% in leave-one-subject-out testing	Two hundred sensing points with explicit cross-subject[130] evaluation
FBG tactile skin	Contact force and location	CNN ³ with multigrid integration	Median errors of 35 mN and 3.2 mm	Curved skin with overlapping FBG ⁴ [7]
Multicore-fiber catheter	Continuous catheter shape	PSO-BP ⁵ neural network	Average errors of 0.57 mm at constant temperature and 0.36 mm at variable temperature	Dedicated temperature-compensation core [9]
Wearable full-body avatar	Continuous joint-position estimates	ANN ⁶ and CNN on an edge processor	3.5 cm average position error and 71% lower power than wireless streaming	On-device inference on a transferable non-fiber strain platform [49]
Soft robotic finger	Closed-loop contact-force control	Stateful LSTM ⁷ and feedback controller	25 Hz control with an average error of 0.17 N	Compensation for drift and hysteresis on a transferable non-fiber soft-sensor platform [71]
Field DAS fault monitoring	Normal or fault decision	Physics-informed generator, denoiser, and classifier	91.8% field accuracy without target-field fault-event training data	Physics-guided generation and transfer across sites [74]

¹ SVM: support vector machine. ² POF: plastic optical fiber. ³ CNN: convolutional neural network. ⁴ FBG: fiber Bragg grating. ⁵ PSO-BP: particle swarm optimization-backpropagation neural network. ⁶ ANN: artificial neural network. ⁷ LSTM: long short-term memory.

Distributed fiber systems create a related machine-to-operation loop. A deployed system may detect an event, estimate its location, classify its severity, and trigger inspection or control. Machine learning has been applied to structural defects [150,151], railway fasteners [13], pipelines, and long-range intrusion monitoring [11]. Evaluation in these settings should report event-level precision and recall, localization error, time to detection, and false alarms over long background recordings. Accuracy on balanced signal clips does not quantify the operational cost of repeated false alarms or missed rare events.

Across soft robots, wearable interfaces, and structural monitoring, system performance depends on the combined dynamics of sensing, inference, and action [29,71,149–151]. Delay, drift, or directional bias may affect stability even when average estimation error is small. Hardware-in-the-loop tests can introduce sensor dropout, communication delay, and unfamiliar inputs before on-body or robotic deployment. Fixed protocols and confidence logging also help separate model improvement from users adapting their movements to the interface.

5. Critical challenges and future horizons

Performance measured on a single dataset may change when the user, device, batch, or operating condition changes [14,16,36–37,50–52]. Deployment also constrains labeling effort, calibration time, memory, latency, and energy. These limitations interact: a compact

model trained on narrow data may be efficient but fragile, while a more expressive model may improve interpolation without improving transfer. The resulting challenges can be traced to three connected parts of the sensing pipeline. Limited experimental coverage motivates physics-guided and label-efficient learning. Shifts between training and deployment require explicit transfer and adaptation tests. Resource limits shift attention to acquisition hardware, compact algorithms, and edge processors. Addressing one constraint may expose another, so data coverage, generalization, and hardware are considered together below.

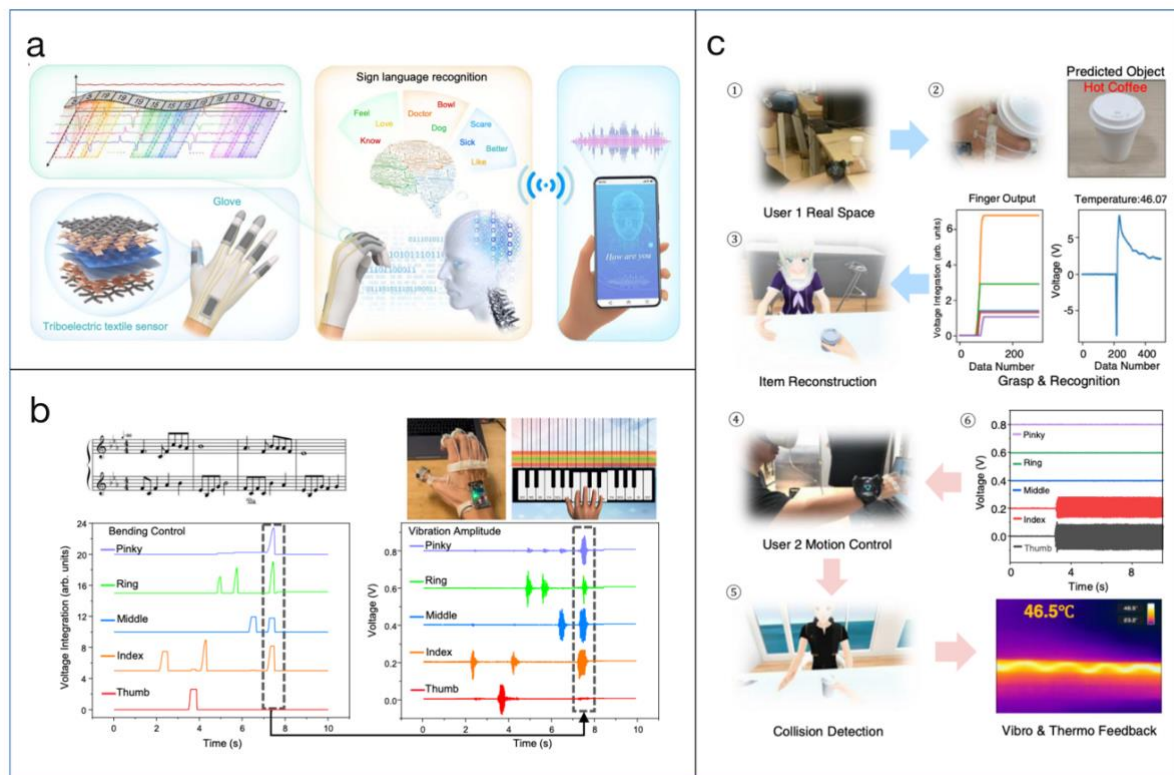


Figure 8. Human-machine interaction and closed-loop haptic feedback: (a) a triboelectric smart glove that combines multichannel gesture signals, AI-based sign-language recognition, and wireless communication (Adapted from Ref. [21]. Copyright 2021, Springer Nature; licensed under CC BY 4.0); (b) simultaneous multi-finger motion control and vibration feedback during virtual-piano interaction (Adapted from Ref. [149]. Copyright 2022, Springer Nature; licensed under CC BY 4.0); and (c) a cross-space interaction loop integrating object recognition, virtual reconstruction, motion control, collision detection, and vibrothermal feedback (Adapted from Ref. [149]. Copyright 2022, Springer Nature; licensed under CC BY 4.0).

5.1. Data scarcity and physics-informed neural networks (PINN)

Training-data coverage remains a practical limitation for learning-based fiber sensors. Experimental datasets are costly and often sample only a subset of devices, users, and operating environments. A distributed trace or wearable recording may yield millions of overlapping windows while representing only one device, participant, installation, or day. Similar concerns have been noted in structural-monitoring and distributed-fiber studies [97,152]. Labels may also depend on calibrated equipment, synchronized references, expert annotation, destructive testing, or naturally rare events [35,153]. Dataset descriptions are therefore more informative when they report devices, batches, participants, sessions, sites, and acquisition days alongside the number of windows.

Physical knowledge can partly compensate when experimental coverage is limited [154]. A physics-informed neural network normally places the residual of a governing differential equation in the loss, but fiber-sensor models can also use transfer functions, finite-element simulations, geometry, monotonicity, conservation relationships, or simulated optical responses. Numerical models of structural monitoring and mechanically designed sensors illustrate how these priors can generate training data or restrict plausible responses [155,156]. Depending on where the prior enters the pipeline, physics guidance acts on the data, architecture, or objective. Electromagnetic, scattering, and mechanical models can generate spectra, distributed traces, strain fields, and contact distributions. Architectures can preserve sensor connectivity, causal order, or invariance to equivalent channel arrangements, while loss functions can enforce force balance, smooth deformation, non-negative pressure, monotonicity, or agreement with an analytical model [154]. A residual model then learns the discrepancy between physical prediction and experiment rather than the complete sensing relation.

At the data level, physics-guided generation is especially relevant when target events are rare, costly, or unsafe to reproduce. For example, Wan et al. combined physically generated DAS target events with measured background data and transferred the resulting framework from a simulation site to a coal-mine conveyor [157]. Its fault-diagnosis accuracy reached 91.8%, compared with 82.2% for a conventional data-driven network and 86.6% for a manually designed algorithm trained on real fault signals. The improvement was obtained for a limited set of events and installations, defining the current boundary of the comparison.

When the physical model is incomplete but unlabeled measurements are abundant, label-efficient learning provides a complementary route. Repeated or paired observations support self-supervised objectives without clean targets [43,59]. In a semi-supervised example, Zhu et al. transferred a phase-picking model from conventional seismic waveforms to DAS and refined it with pseudo-labels [158]. Self-supervision draws information from measurement structure, whereas semi-supervision relies on a transferable model, a labeled subset, or sufficiently reliable pseudo-labels.

Neither route removes model error. Inaccurate constitutive laws, boundary conditions, coupling assumptions, or simulated noise can bias physics-guided learning, while unreliable pseudo-labels can reinforce errors in semi-supervised models. Comparisons with data-only and physics-only baselines, together with ablations of simulated data, experimental data, architectural constraints, and physics-based losses, help locate the source of improvement [154]. Uncertainty estimates are also needed when neither the physical model nor the training data adequately cover the input [159].

Physics-guided, self-supervised, semi-supervised, and active-learning methods therefore address different forms of missing coverage. Governing models supply simulation, repeated measurements supply self-supervision, and partial labels guide targeted sampling. Their contribution lies in broadening physical and experimental coverage rather than multiplying overlapping windows. The remaining challenge is to recognize conditions unsupported by both the data and the assumed physics [154,159].

5.2. Model generalizability and domain adaptation

Broader training coverage does not remove the systematic shifts introduced by a new user, device, or time point. Anatomy, motion style, garment fit, strain transfer, mounting, material batch, and ageing can change both the signal distribution and its relationship to the measurand. Transfer learning and domain adaptation reuse shared representations while updating a limited part of the model [160,161]. Feature alignment, adversarial removal of device identity, meta-learning, and test-time adaptation target different components of this shift. Claims of transfer depend on how these domains are separated

during validation. Participant-wise splits test new users, device- and batch-wise splits test hardware variation, and chronological splits probe ageing or repeated use. Passing one test can still conceal failure on another. A model may generalize to a new user wearing the original garment yet fail when both user and garment change. A domain matrix across users, devices, batches, and time makes this boundary explicit.

When relevant data remain distributed across users, devices, or institutions, federated learning can train a shared model without centralizing raw physiological measurements [162,163]. This arrangement may suit smart clothing and clinical wearables, but it does not automatically provide privacy or consistent performance. Model updates may leak information, uneven local datasets can bias training, and communication consumes energy. Personalization and continual learning can adapt the shared model, while safety-relevant updates still require verification before replacing a validated version.

At deployment, domain adaptation often appears as a calibration problem. For example, Fasseaux et al. and Jiao et al. applied learned calibration to plasmonic-fiber biosensing and WDM FBG demodulation, respectively [40,41,44,62,65]. The resulting mappings nevertheless remain tied to their training devices and conditions. Reducing calibration effort is therefore more realistic than assuming calibration can be eliminated. Uncertainty may trigger a new reference measurement when the domain changes, whereas unstable contact, poor placement, or excessive noise may require repositioning or rejection. Distinguishing these causes is essential before adaptation is automated.

5.3. Edge computing and hardware-friendly algorithms

Smart clothing, prosthetics, soft robots, and wireless monitoring devices are often constrained by memory, processing power, communication bandwidth, and battery capacity during operation. Even a transferable model has to fit the computational and energy budget of the complete sensing node. Besides, continuous cloud streaming may also conflict with latency, connectivity, or privacy requirements. Wearable and integrated-fiber architectures therefore place part of the computation near the sensing material [5,8,49,164,165]. Model compression is one route to this local computation. Compact architectures, pruning, low-bit quantization, knowledge distillation, and low-rank parameterization reduce arithmetic and memory demand. For example, Loke et al. embedded memory for a 1,650-connection neural network in a digital fiber and classified four activities from 12 s temperature samples with 96.4% test accuracy after offline training [101]. The in-fabric implementation integrates sensing, model storage, and inference, while power, updating, and long-term reliability remain separate engineering questions.

Compression alone does not address the cost of acquiring and transmitting redundant data. Adaptive sampling and event-triggered acquisition suppress inactive periods, while channel selection reduces dense-array dimensionality. In a related event-driven setting, Taunyazov et al. processed asynchronous visual and tactile changes rather than continuously sampled frames [147,166–168]. The acquisition principle is transferable, although its energy advantage has not yet been quantified on a fiber or textile platform. These choices determine how computation is divided among the sensing fiber, a nearby processor, and a remote service. Immediate detection and feedback can remain local, while model updates and longitudinal analysis run less frequently on more capable hardware. Smart-textile architectures already partition sensing, interconnection, computation, and feedback in this way [70]. Their long-term performance will also depend on interconnect reliability, thermal management, mechanical durability, washability, and repeated deformation.

Practical comparisons of edge systems therefore pair sensing performance with computational cost. Processor type, numerical precision, memory, on-device latency, operating power, communication demand, and calibration stability describe whether local inference can be sustained [101]. Compact recalibration and privacy-preserving adaptation become additional requirements as users, batches, and material properties change. These system-level measures connect algorithmic efficiency to the conditions of continuous use.

6. Conclusion

Machine learning is most useful in fiber sensing when it is integrated with the physical and computational sensing pathway. Sensor architecture determines which states are observable, signal models address noise and history dependence, and temporal, spatial, or multimodal representations support application-specific estimates and actions. Coordinating these layers provides a route from responsive fibers to sensing systems that can interpret their operating state.

The main unresolved issue is whether these capabilities persist across devices, fabrication batches, users, and operating conditions. Physics-constrained models, independent-device validation, manufacturing metadata, and uncertainty-aware abstention may improve transfer, while wearable and embedded platforms also require latency, memory, energy, and calibration reporting. Reproducible operation across these boundaries would provide a stronger basis for deployment than peak accuracy from a single device or dataset.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
AM	Additive manufacturing
ANN	Artificial neural network
CNN	Convolutional neural network
DAS	Distributed acoustic sensing
EHD	Electrohydrodynamic
EMD	Empirical mode decomposition
FBG	Fiber Bragg grating
FEM	Finite element method
HMI	Human–machine interaction
LSTM	Long short-term memory
ML	Machine learning

PCF	Photonic crystal fiber
PINN	Physics-informed neural network
POF	Plastic optical fiber
PSO-BP	Particle swarm optimization–backpropagation
SNR	Signal-to-noise ratio
SURF	Speeded-up robust features
SVM	Support vector machine
VMD	Variational mode decomposition
WDM	Wavelength-division multiplexing

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